

Demand and supply effects of university funding changes

An Australian policy analysis

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Motivation and Research Question

Motivation

- ▶ University enrolments are key to understanding the future labour market
- ▶ What policy tools are available to divert higher education students?

Three key questions:

1. Are university applicants responsive to price changes?
2. Do universities exhibit revenue-maximising behaviour?
3. What are the equity impacts of differential pricing policies?

The Policy Change: Job-ready Graduates

 The Guardian

Australian university fees to double for some arts courses, but fall for Stem subjects

The education minister, Dan Tehan, will announce the policy to create more “job-ready graduates” at the National Press Club on Friday...

18 June 2020



 The Conversation

Fee cuts for nursing and teaching but big hikes for law and humanities in package expanding university places

Under the plan to produce “job-ready graduates”, to be outlined on Friday by Education Minister Dan Tehan, those opting for teaching,...

18 June 2020



 ABC

Humanities degrees set to double in price as Parliament passes higher education bill

legislation will provide more university places for Australian students, make it cheaper to study in areas of expected job growth and provide...

7 Oct 2020



Job-ready Graduates Funding Changes

Policy Change	Price change (student)	Revenue change (university)	Fields of Study
Cheaper & more funding	-17% -41%	4% 34%	IT, Architecture, Public Health Language
Cheaper & less funding*	-59% to -41% -29% to -17% -29% to -16%	-15% to -4% -31% to -28% -15% to -1%	Nursing, Teaching, Maths, Agriculture Medical Science, Environments Science, Engineering, Pharmacy, Optometry, Allied Health, Health (Broad)
More expensive & more funding*	1% to 19% 30% 51% to 117%	6% to 21% 17% 9% to 22%	Psychology, Medicine, Dentistry, Veterinary, Social Work, Creative Arts Business, Law, Economics Arts (Broad), Criminology
More expensive & less funding	67% 117%	-8% -11% to -22%	Creative Arts (Broad) Politics, History, Communications, Human Movement

Job-ready Graduates: Government Claims

“We are sending a **price signal** to encourage people to study in areas of **expected employment growth**”



“We think that the **price incentives** and the price changes that we’ve made, that students will pay attention to them, and **will focus on them**”

University funding in Australia

Per-student university revenue is given by

$$\text{Revenue} = \underbrace{\text{Government Contribution}}_{\text{CSP Subsidy}} + \underbrace{\text{Student Contribution}}_{\text{HECS-HELP Loan}}$$

where:

- ▶ Commonwealth Supported Places (CSPs) are generally available to all Australian citizens
- ▶ HECS-HELP is an income contingent loan
 - ▶ Repayments begin from \$48K (annual income)

Literature Review

There are **no** published papers that empirically analyse the impacts of the Jobready Graduates Legislation.

My literature review mainly consists of:

- ▶ Theoretical papers on Jobready Graduates (published before the legislation passed)
 - ▶ (Chapman and Khemka 2022) and (Daly and Lewis 2020)
- ▶ Empirical papers on university price changes
 1. In Australia's history: (Aungles et al. 2002), (Deloitte Access Economics 2011)
 2. In other countries: (Coelli 2009), (Neill 2009)

Data

Unit-record data from Universities Admissions Centre (UAC):

- ▶ Undergraduate (Bachelor), domestic, CSP applicants
 - ▶ Background characteristics: gender, indigenous, SES, age, school etc.
- ▶ 2014 to 2022
 - ▶ 7 years pre-policy; 2 years post-policy
- ▶ Preferences and subsequent enrolments
 - ▶ Applicants: $n=736,026$
 - ▶ Enrolments: $n=371,890$

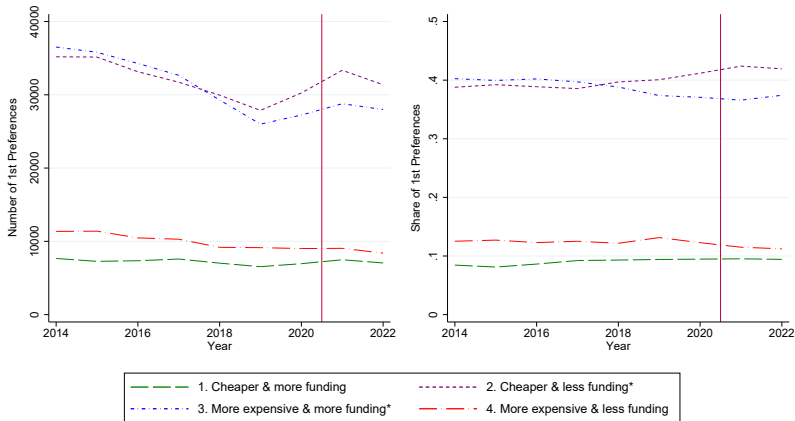
Data: Fictitious applicant data

Year	Age	Gender	Leaver	Indigenous	SES	Postcode	Language	School	ATAR
2016	18	F	Yes	No	High	2043	English	Private	90-95

Preference	Course ID	Field of study	Course Start	Uni group	Level	Enrol?
1	5492	010101	Feb-16	Go8	Bachelor	No
2	5978	020101	Feb-16	Go8	Bachelor	Yes
3	122	020300	Mar-16	Metro	Bachelor	No
4	1679	060100	Feb-16	Metro	Bachelor	No
5	4189	010101	Feb-16	Regional	Bachelor	No

Note: ‘Leaver’ is a dummy variable for a high school applicant, and ‘Language’ captures the applicant’s primary language spoken at home. ‘School’ specifies whether the applicant attended a private or public high school. ATAR is given as a range, which is finer at higher ATARs. Age is calculated from year of birth data. SES is calculated from self-reported parental educational attainment data.

Demand: 1st Preferences for field of study



Demand-side Theory

Representative 18-year-old university applicant's lifetime utility function (conditional on applying to university):

$$U_i(f_v) = \underbrace{\sum_{t=19}^{65} \frac{y_{it}(f_v)}{(1+r_i)^{t-19}}}_{\text{PV Income}} - \underbrace{\sum_{t=19}^{65} \frac{c_{it}(f_v, y_{it}(f_v))}{(1+r_i)^{t-19}}}_{\text{PV Costs}} + \underbrace{\gamma_i(f_v)}_{\text{Preferences}}$$

► where income y and costs c are for field-of-study f at time t

Assuming they maximise lifetime utility:

$$\max_{f_v} U_i(f_v) \quad \text{subject to} \quad ATAR_i \geq \underline{ATAR}_{f_v}$$

Demand-side Results

$$U_j^* = \underbrace{W'_j \beta_j}_{\text{Student-specific}} + \underbrace{X'_j \gamma}_{\text{Course-specific}} + \epsilon_j$$

	(1) All Applicants	(2) Low SES	(3) Medium SES	(4) High SES
log(Student Contribution)	-0.024 (0.036)	-0.015 (0.054)	-0.033 (0.030)	-0.031 (0.049)
log(University Revenue)	0.008 (0.102)	0.054 (0.163)	-0.090 (0.089)	-0.021 (0.133)
% Random Sample	10%	50%	50%	10%
N cases	71,672	33,294	100,478	39,544

	(5) School Leaver	(6) Non-leaver	(7) English	(8) Non-English
log(Student Contribution)	-0.028 (0.047)	-0.038 (0.041)	-0.002 (0.041)	-0.040 (0.033)
log(University Revenue)	0.195 (0.132)	-0.045 (0.118)	0.006 (0.120)	0.150 (0.085)
% Random Sample	10%	20%	10%	55%
N cases	35,911	71,454	51,218	113,231

Demand-side Results: Elasticities

Re-specified model: 4 choices instead of 27

► Price ↑ / ↓ and revenue ↑ / ↓

Model Specification	Student Price Elasticity	t-stat	n
Low SES	0.0216	0.66	56,565
Med SES	-0.0192	-1.08	173,321
High SES	-0.0576***	-4.76	349,966
Leaver	-0.0473***	-3.96	336,038
Non-leaver	-0.0211	-1.41	292,464
English	-0.0296**	-2.82	448,129
Non-English	-0.0726***	-3.72	180,373
All Applicants	-0.0395***	-4.28	628,502

† $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Note: The reported elasticity is the weighted average of the model's four calculated elasticities, weighted by enrolments. There are missing observations for SES, so not all applicants fall into the three categories reported.

Demand-side Results: Marginal Effects

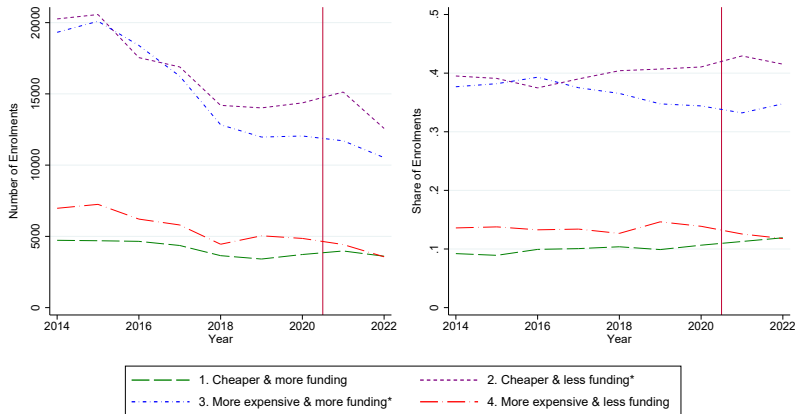
Funding Change	<i>Change in</i>		<i>Share of preferences</i>		<i>%Δ in shares</i>	
	Price	Revenue	2020	2021	Observed	Model
1. Price ↓ and Revenue ↑	-17%	3%	10.7%	11.3%	5.92%	0.80%
2. Price ↓ and Revenue ↓*	-26%	-13%	41.1%	42.9%	4.58%	0.88%
3. Price ↑ and Revenue ↑*	27%	14%	34.4%	33.2%	-3.47%	-1.19%
4. Price ↑ and Revenue ↓	116%	-7%	13.9%	12.6%	-9.48%	-3.95%

Note: *Misaligned policy because the student and university incentives are misaligned. Within each broad category shown, there are 4-11 unique fields of study. To calculate the funding change for each broad category, I take the average of each field-of-study funding change, weighted by 2020 enrolment levels.

Key:

1. IT, Architecture, Public Health, and Language
2. Nursing, Teaching, Maths, Agriculture, Science, Engineering, and Health
3. Business, Law, Psychology, Medicine, Social Work, and Creative Arts
4. Politics, History, Communications, and Human Movement

Supply: Enrolments



Supply-side Theory

- University net teaching revenue (NTR) for course c and university v at time t is:

$$NTR_{cvt} = \underbrace{\sum_{c=1}^C G_{ct} E_{cvt} + \sum_{c=1}^C S_{ct} E_{cvt}}_{\text{Revenue}} - \underbrace{\sum_{c=1}^C (FC_{cvt} + MC_{cvt} E_{cvt} + MC_{cvt}^2 E_{cvt}^2)}_{\text{Costs}}$$

- If they are net-revenue-maximisers:

$$\max_{E_{cvt}} NTR_{cvt} \quad \text{subject to}$$

$$\underbrace{\sum_{c=1}^C G_{ct} E_{cvt} \leq \bar{G}_{vt}}_{\text{MBGA limit}} \quad \text{and} \quad \underbrace{E_{cvt} \leq \min(\bar{E}_{cvt}^{\text{Applicants}}, \bar{E}_{cvt}^{\text{Supply}})}_{\text{Enrolment constraints}}$$

Supply-side Estimation

Analyse how enrolments change with funding:

$$\log(Enrolments_{fgt}) = \beta_0 + \beta_1 \log(Revenue_{fgt}) + \eta_{fg} + \gamma_{gt} + u_{fgt}$$

- ▶ η_{fg} is a field-of-study and university-group fixed effect
- ▶ γ_{gt} is a university-group and time fixed effect

Supply-side Results (1/2)

	Dependent Variable: Log Enrolments			
	(1)	(2)	(3)	(4)
log(Revenue)	-0.25 (0.17)			
log(Revenue) × Go8		-0.56 [†] (0.30)		
log(Revenue) × Metro		-0.45 (0.28)		
log(Revenue) × Regional		0.14 (0.30)		
log(Revenue) × Other		0.36 (0.57)		
log(Student Contribution)			-0.043 (0.097)	
log(Government Contribution)			0.034 (0.046)	
University Group × Field of Study FE	✓	✓	✓	
University Group × Year FE	✓	✓	✓	
Constant	5.04** (1.69)	4.79* (1.75)	2.69* (1.27)	
Observations	10,579	10,579	10,579	
Adjusted R^2	0.336	0.336	0.336	

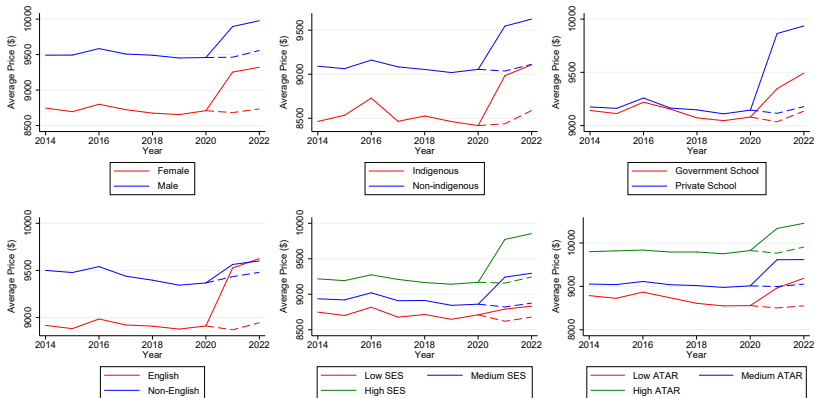
[†] $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Note: Revenue is the sum of the student contribution and the government contribution. All dollar amounts are in real terms, indexed to 2022. Standard errors are clustered at the broad 27-group field of study level because that is the expected level of correlation.

Supply-side Results (2/2)

	Dependent Variable: Log Enrolments		
	(1)	(2)	(3)
$\log(\text{Student Contribution}) \times \text{Go8}$			-0.046 (0.21)
$\log(\text{Student Contribution}) \times \text{Metro}$			-0.27 (0.22)
$\log(\text{Student Contribution}) \times \text{Regional}$			0.090 (0.17)
$\log(\text{Student Contribution}) \times \text{Other}$			0.51 (0.55)
$\log(\text{Government Contribution}) \times \text{Go8}$			0.059 (0.093)
$\log(\text{Government Contribution}) \times \text{Metro}$			-0.13 (0.093)
$\log(\text{Government Contribution}) \times \text{Regional}$			0.14 (0.098)
$\log(\text{Government Contribution}) \times \text{Other}$			0.36 [†] (0.18)
University Group $\times$ Field of Study FE			✓
University Group $\times$ Year FE			✓
Constant			2.73* (1.23)
Observations			10,579
Adjusted R^2			0.336

[†] $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. *Note:* Revenue is the sum of the student contribution and the government contribution. All dollar amounts are in real terms, indexed to 2022. Standard errors are clustered at the broad 27-group field of study level because that is the expected level of correlation.

Cost Effects



Note: The dashed line represents a counterfactual scenario if JRG funding changes did not happen.

Conclusions

Key findings:

1. University applicants are **not** particularly price sensitive
2. Universities **do not** exhibit overt revenue-maximising behaviour
3. JRG disproportionately affected women, indigenous, private school, and high SES applicants

Policy-relevant implication:

- ▶ If the government wants to have control over the future labour market, it should not use price as an incentive for students, and maybe not for universities either.

Did it stack up?

- ✓ “We are sending a **price signal**”
- ? “Encouragement to study in areas of **expected employment growth**”
- ✗ “Students will pay attention to and focus on **price incentives**”



Questions/concerns/feedback/criticisms

All feedback is welcome!

Feel free to email me at moyong@student.unimelb.edu.au